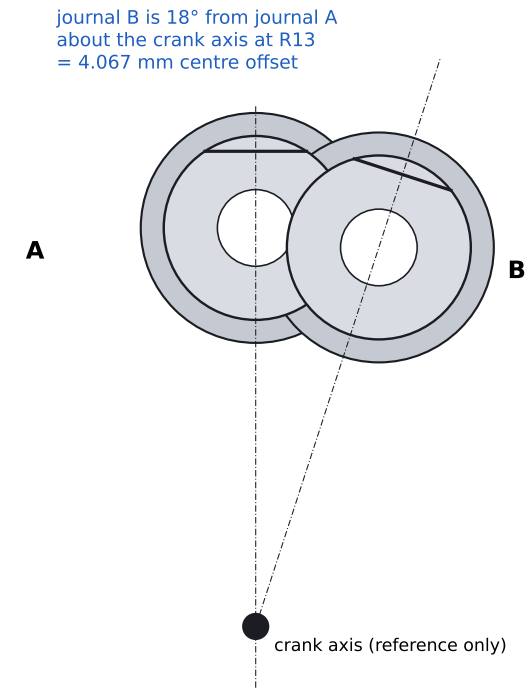
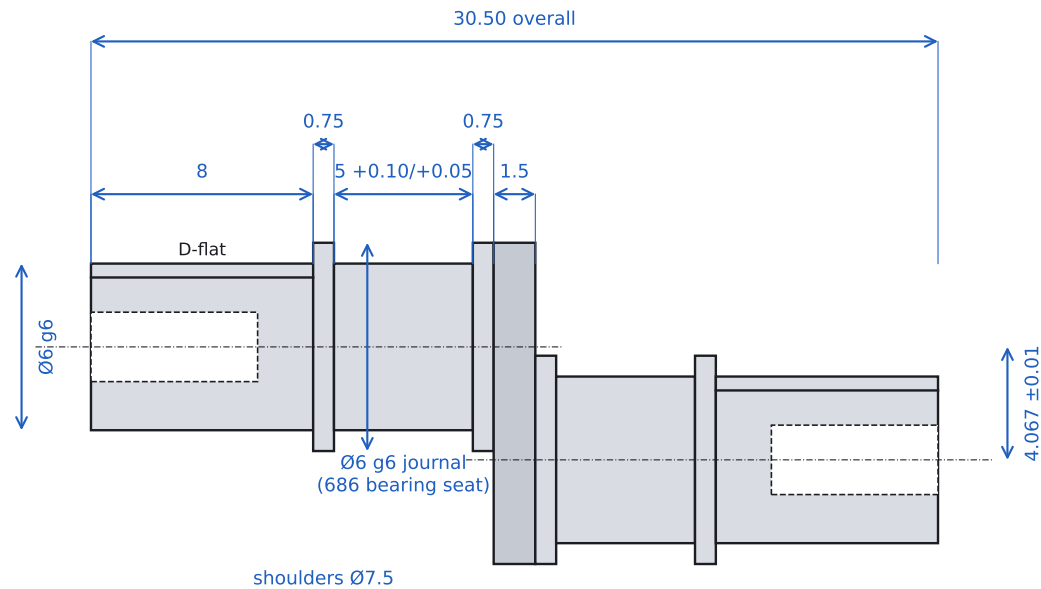


# M01 Split crankpin

Material: Stainless 303 / 416 (or 1144 steel, black oxide) Qty: 5 per engine / 250 for 50 units Units: mm V10 display engine (original design)

SIDE VIEW (end A = front journal)

END VIEW from end A

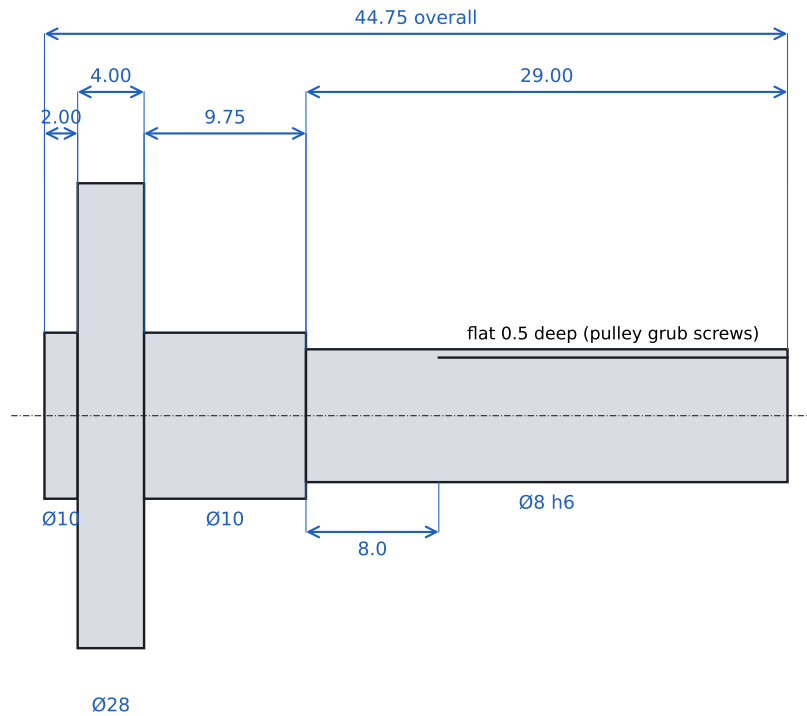


- Both journals Ø6 g6 (-0.004/-0.012), Ra 0.4 or better - they carry the 686 bearing inner rings.
- Journal centre offset 4.067 ±0.01 at 18° about the crank axis (R13). All other dims ±0.05.
- D-flats 0.5 deep, each facing radially AWAY from the crank axis (see end view). Flats key the pin into the printed webs.
- Both ends tapped M3 x 6 deep (tap drill Ø2.5). Break all edges 0.2. Shoulder faces square to axis within 0.02.
- Journal B side mirrors journal A (same lengths). The side view is drawn in the plane of the offset.

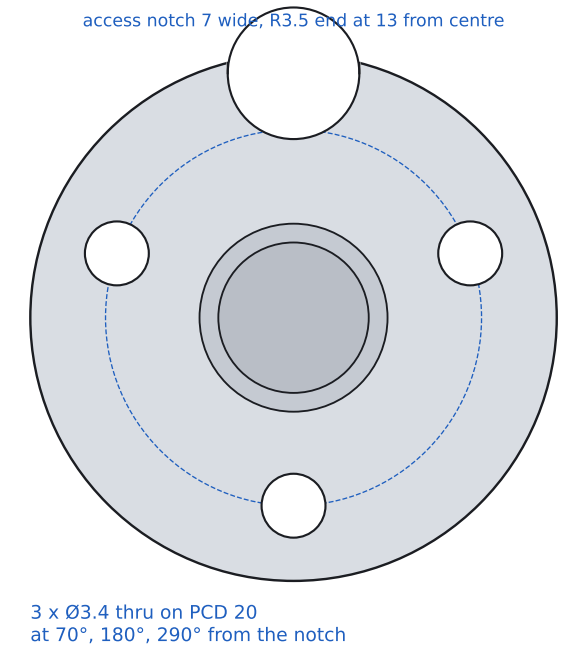
## M02 Main shaft (front and rear identical)

Material: Stainless 303 (or 1045, black oxide) Qty: 2 per engine / 100 for 50 units Units: mm V10 display engine (original design)

SIDE VIEW



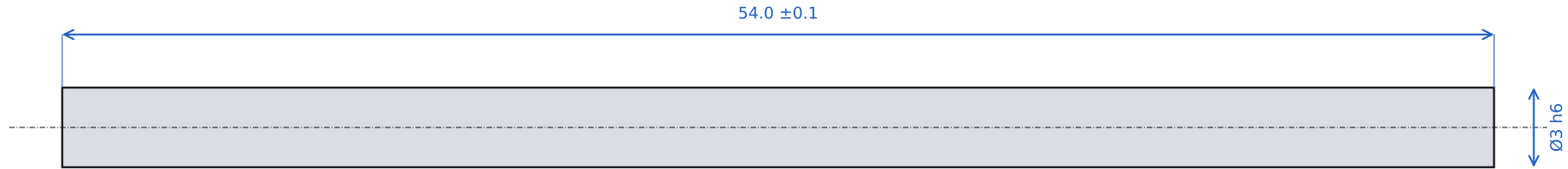
FLANGE END VIEW



- Ø8 journal h6 (0/-0.009) for the 608 bearing; Ra 0.8. Shoulder face square to axis within 0.02 (it locates the bearing inner ring).
- Flange face square to axis within 0.02; spigot Ø10 -0.02/-0.05 (locates in the printed end web). Other dims ±0.05.
- Flange holes are clearance holes; the UNEVEN bolt pattern (110/110/140° spacing) plus the notch makes the shaft fit the web one way only.
- The flat is only needed on the FRONT shaft (drive pulley). Machining it on both keeps one part number.
- Alternative if machining capacity is short: 8 mm h6 ground rod + commercial 8 mm rigid flange coupling (lower precision; not recommended for production).

## M03 Piston guide rail

Material: 3 mm ground stainless rod (h6) Qty: 10 per engine / 500 for 50 units



- Cut from 3 mm h6 ground stainless linear rod (not drill rod - needs corrosion resistance and a polished surface).
- Both ends:  $0.3 \times 45^\circ$  chamfer, deburred so they do not scratch the bronze bushings. Length  $\pm 0.1$ .
- The rail is the ONLY thing the piston slides on (two oil-impregnated bronze bushings). Keep it clean; one drop of light oil at assembly.